Manuel Zechmann

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Education

Universitat Pompeu Fabra (UPF) — Barcelona, Spain
PhD in Information and Communication Technologies — *Oct 2026 – Present*

- Research focus: Path Tracing

University of Vienna — Vienna, Austria
Master of Science (MSc), Media Computer Science, Grade Avg: 1.58 (1 = best) — *Mar 2023 – Nov 2025*

- Master's Thesis: Comparative Analysis of Procedural Planet Generators (GAME-ON 2025). Investigated algorithmic terrain synthesis, LOD optimization, and real-time rendering performance.
- Coursework: Advanced computer graphics, image synthesis, path tracing, computer vision, machine learning, and mathematical foundations of rendering.
- Led a 4-person team in a microservice architecture software engineering project.

University of Vienna — Vienna, Austria
Bachelor of Science (BSc), Media Computer Science, Grade Avg: 2.32 (1 = best) — *Oct 2019 – Jan 2023*

- Bachelor's Thesis: Training a CNN to Detect Hands and Handposes. Designed a procedural data generation pipeline in Unity to train robust deep learning models.
- Coursework: Foundations in algorithms, data structures, Computer Vision, and Computer Graphics.

Publications

Comparative Analysis of Procedural Planet Generators — 2025
Manuel Zechmann, *Helmut Hlavacs* — *GAME-ON 2025, Ghent*

- Published and presented research paper focusing on procedural planet generation and performance metrics.

Awards

UNIVIE Research Award for Students
University of Vienna, 2025

- Awarded for academic research and publication of the paper "Comparative Analysis of Procedural Planet Generators".

Leistungsstipendium
University of Vienna, 2025

- Received the Leistungsstipendium (Merit Scholarship) for top-tier academic performance.

Experience

Snap Inc. (Specs / Spectacles) — Vienna, Austria
Research Intern – Computer Vision — *Aug 2025 – May 2026*

- Built synthetic 3D data generation systems using procedural generation, 3D Gaussian Splatting, VLMs, and agentic workflows to create training data for computer vision models. Developed augmentation methods to reduce the sim-to-real gap.
- Trained and adapted vision models for 3D scene understanding, proving effectiveness of generated data with ablation studies.
- Engineered large-scale data generation pipelines on GCP (Vertex AI, GCS), generating up to 5 TB of structured data to train robust vision models.
- Attended and led reading groups, worked in cross-team projects, and proactively supported coworkers, e.g. by supervising recording sessions.

ViewApp — Vienna, Austria
Multimedia Software Engineer — *Jul 2024 – Sep 2024, Jan 2025 – Jul 2025*

- Engineered custom C++ Unreal Engine tools and world-building pipelines, implemented snapping mechanics, and helped ship an Unreal Engine game to Steam, PlayStation 5, and Xbox.
- Led the technical implementation of the core interactive tutorial system, coordinating necessary cross-functional deliverables.
- Developed workflow automation tools and configured automated build processes using TeamCity to optimize production efficiency.

Partium Vienna, Austria
Junior Software Engineer *Oct 2022 – Mar 2024*

- Trained computer vision models for spare part recognition and integrated OCR pipelines for high-precision text extraction.
- Managed containerized microservices using Docker and Kubernetes in beta testing environments.

Partium Vienna, Austria
Software Engineer Intern *Jul 2022 – Sep 2022*

- Evaluated and integrated Elasticsearch as a high-performance search backend, accelerating large-scale dataset retrieval.

Projects

Path Guiding | Path Tracing, C++ Oct 2026

- Implemented Practical Path Guiding (2017) algorithm, reducing MSE and FLIP metrics by up to 27% and 28% respectively at equal render time compared to my path tracer with NEE and MIS.

VulkanParticles Game Engine | C++, Vulkan, CMake, GoogleTest May 2026 – Present

- Architected and developed a high-performance C++ particle game engine utilizing the Vulkan API, designed for modular expansion into a full-scale engine.
- Implemented robust software engineering practices, featuring automated CI/CD pipelines and comprehensive unit testing with GoogleTest.

GPT from Scratch (nanoGPT) | Python, PyTorch, Transformers 2024

- Engineered a decoder-only transformer architecture from scratch in Python and PyTorch, based on Andrej Karpathy's reference implementation.
- Trained and evaluated the language model, gaining hands-on expertise in tokenization, self-attention mechanisms, and compute optimization.

Tall Trunk Galaxy 64 (ROM Hack) | C, Blender, SM64 SDK, Procedural Design Apr 2024

- Developed a small SM64 game utilizing the C source code and Blender asset workflows within strict N64 hardware constraints.

Skills

Languages: C++, Python, C, SQL, GDScript
Computer Graphics: Path Tracing / Monte Carlo rendering, Physically Based Rendering, Vulkan, GLSL/HLSL, CUDA (learning), Unreal Engine, Procedural Generation, 3D Gaussian Splatting, Blender, NVIDIA PhysX
Computer Vision: PyTorch, 3D Scene Understanding, Synthetic Data Generation, Sim-to-Real, Ablation Studies, VLMs, ControlNet, Agentic Workflows (MCP)
Software Engineering: Profiling (Nsight, RenderDoc), Multi-threading, CI/CD, Git, Docker, Kubernetes, GCP, GoogleTest
Research Interests: Real-Time / Neural / Physically Based Rendering, XR, 3D Scene Understanding, AI
Spoken Languages: German (Native), English (C2), Spanish (A2)